Colimits of Higher Categories

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Any environment worthy of the name "category" should somehow include a theory of limits and colimits. Having spent some time discussing higher category theory and some of its features, we will today discuss the notion of a colimit in the context of higher categories. The main goal is to discuss the abstract theory of colimits for higher categories, rather than focusing on computability of the definition. An important source for the concept of colimits is clearly the book by Lurie [Lu09]. In particular subsection 1.2.12 and 1.2.13. As the theory of limits and colimits are dual to each other, we will solely focus on colimits here.

Background: Colimits in Ordinary Categories

Before we actually start let us recall the case in ordinary categories. Recall that colimits and limits are a fundamental tools in ordinary category theory, which allow us to build new objects with some universal properties out of old ones. There are dozens of equivalent definitions, one of the most common being the following.

Definition 1.1. For a functor $F\colon J \to \mathcal{C}$, $\operatorname{colim} F$ consists of an object of $\mathcal{C}$, generally written as "$\operatorname{colim} F$", along with a cone $\tau\colon F \to \operatorname{colim} F$ such that every other cone $F \to c$ factors uniquely though τ.

Remark 1.2. Although it is common to write $\operatorname{colim} F$ for the associated object of $\mathcal{C}$, the cone τ is a necessary piece of information for the structure of $\operatorname{colim} F$ as a colimit.

Thus it might be better to rephrase our definition that includes all relevant pieces of information. For that we first need some basic definitions.

Definition 1.3. Let $\mathcal{C}$ and $\mathcal{D}$ be two categories. We define the *join* of two categories, $\mathcal{C} \star \mathcal{D}$, as follows:

Obj Objects are just objects of $\mathcal{C}$ and $\mathcal{D}$.

Mor For two objects x, y we have the following four cases:

$$Homa_{\mathcal{C}\star\mathcal{D}}(x,y) = \begin{cases} Hom_{\mathcal{C}}(x,y) & \text{if } x,y \in \mathcal{C} \\ Hom_{\mathcal{D}}(x,y) & \text{if } x,y \in \mathcal{D} \\ * & \text{if } x \in \mathcal{C},\ y \in \mathcal{D} \\ \emptyset & \text{if } x \in \mathcal{D},\ y \in \mathcal{C} \end{cases}$$

Remark 1.4. Essentially, we are taking the categories $\mathcal{C}$ and $\mathcal{D}$ and then add a unique arrow from each object in $\mathcal{C}$ to each object in $\mathcal{D}$.

We have following important special case.

Example 1.5. Let $\mathcal{D} = [0]$, the unique category with one object and one identity morphism. Then we denote

$$\mathcal{C} \star \mathcal{D} = \mathcal{C}^{\triangleright}$$

and call it the *cone on* $\mathcal{C}$ We normally denote the uniqe object coming from $\mathcal{D}$ as v and call it the *vertex point*. Note that it is the final object of the category $\mathcal{C}^{\triangleright}$.

Having a notion of a cone we can make following definition

Definition 1.6. Let $F : J \to \mathcal{C}$ be a functor. We define the category of cones, denoted by $\mathcal{C}_{F/}$, as the category which as follows:

Obj Functors $\tilde{F} : J^{\triangleright} \to \mathcal{C}$, such that $\tilde{F}|_J = F$, called *cones over* F.

Mor For two objects $\tilde{F}, \hat{F}$ we have natural transformations $\alpha : \tilde{F} \to \hat{F}$, such that $\alpha|_J$ is the identity.

Remark 1.7. Essentially this category has as objects cones over the diagram F and as maps and a map between the vertex points which commutes with the cones.

Now, we give our definitions for a colimit in two steps:

Definition 1.8. We say an object i is *initial* if for any other object $c \in \mathcal{C}$, we have

$$Hom_{\mathcal{C}}(i,c) = *$$

i.e. there exists a unique map from i to any other object.

Definition 1.9. For a functor $F\colon J \to \mathcal{C}$, we define $\operatorname{colim} F$ as the initial object in the category $\mathcal{C}_{F/}$ of cones under F.

Our goal for the coming sections is to generalize these constructions to the level of quasi-categories. In order to achieve that we need to generalize the notions of join, slices.

Join and Slice

In the last section we defined the join of two categories and how we can use that to define the category of cones. Now, a functor $F\colon I \star J \to \mathcal{C}$ consists of two functors $F|_I\colon I \to \mathcal{C}$ and $F|_J\colon J \to \mathcal{C}$ along with cones $F|_I \to F|_J(j)$ varying naturally in $i \in I$. This is exactly the data of a functor $J \to \mathcal{C}_{F|_I/}$. In other words, we can regard $I \star J$ as an object of $\mathcal{C}\mathrm{at}_{I/}$, and we get an adjunction,

$$\mathcal{C}\mathrm{at} \underset{\mathcal{C}_{-/}}{\overset{(I \to I \star -)}{\rightleftarrows}} \mathcal{C}\mathrm{at}_{I/}$$

which comes from an equivalence of maps

$$\frac{K \mapsto (I \to I \star K)}{(p\colon I \to \mathcal{C}) \mapsto \mathcal{C}_{p/}}$$

It turns out that this can be generalized to simplicial sets. We will not define the join in detail, but rather focus on important features we need in order to later discuss colimits.

Fact 2.1. For two simplicial sets S, T, there is a *join* $S \star T$.

For a detailed definition of the join construction see 1.2.8 of [Lu09].

Definition 2.2. We define the right cone,denoted as $S^{\triangleright}$ as $S \star \Delta^0$.

Definition 2.3. For a simplicial map $p\colon S \to \mathcal{C}$, there is a *slice category* $\mathcal{C}_{p/}$ that models cones under p. It can be defined by using the universal property of the following adjunction,

$$s\mathcal{S}\mathrm{et} \underset{\mathcal{C}_{-/}}{\overset{(S \to S \star -)}{\rightleftarrows}} s\mathcal{S}\mathrm{et}_{S/}$$

which comes from an equivalence of maps

$$\frac{T \mapsto (S \to S \star T)}{(p\colon S \to \mathcal{C}) \mapsto \mathcal{C}_{p/}}$$

Remark 2.4. Note that $(\mathcal{C}_{p/})_0$ consists of maps $S^{\triangleright} \to \mathcal{C}$ that extends p.

We have following crucial facts about the join-slice construction.

Fact 2.5.

- Given $j\colon T \to S$ and $p\colon S \to \mathcal{C}$, restriction gives a map $\mathcal{C}_{p/} \to \mathcal{C}_{pj/}$.

- If j is injective, then $\mathcal{C}_{p/} \to \mathcal{C}_{pj/}$ is a left fibration. In particular, the projection $\mathcal{C}_{p/} \to \mathcal{C}$ is a left fibration.
- Thus, if $\mathcal{C}$ is a quasicategory, then so too is $\mathcal{C}_{p/}$.
- If the following commutative square

$$\begin{tikzcd} S_0 \arrow[r] \arrow[d] & S_1 \arrow[d] \\ S_2 \arrow[r] & S_{12} \end{tikzcd}$$

  is a pushout diagram of simplicial sets such that there are maps $p_i \colon S_i \to \mathcal{C}$. Then the induced diagram

$$\begin{tikzcd} \mathcal{C}_{p_{12}/} \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & \mathcal{C}_{p_1/} \arrow[d] \\ \mathcal{C}_{p_2/} \arrow[r] & \mathcal{C}_{p_0/} \end{tikzcd}$$

  is a pullback diagram.

For some of the proofs (and some more definitions) on categories under maps see 1.2.9 in [Lu09].

Example 2.6. Consider the simplicial set Λ^2_0,

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & 2 \\ 1 & \end{tikzcd} .$$

Then the right cone $(\Lambda^2_0)^{\triangleright}$ is given by adjoining a vertex,

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] & 2 \\ 1 & v \end{tikzcd} ,$$

and freely adjoining paths to the vertex, along with homotopies of paths and so forth,

$$\begin{tikzcd} 0 \arrow[r] \arrow[d] \arrow[dr] & 2 \arrow[d] \\ 1 \arrow[r] & v \end{tikzcd} .$$

So, $(\Lambda^2_0)^{\triangleright} \cong \Delta^1 \times \Delta^1$.

Remark 2.7. Note there is nothing fundamentally difficult with constructing the join. We can think of it as extending addition on the category Δ.

Initial Objects

Recall our goal was to define colimits as initial objects in an appropriate category. It is now not difficult to define initial objects. In the first section we gave a definition for initial objects that will readily generalize to the contex of quasi-categories.

Definition 3.1. We say an object $c \in \mathcal{C}$ is *initial* if for any other object $d \in \mathcal{C}$ the mapping *space*

$$map_{/\mathcal{C}}(c, d)$$

is contractible

While this definition works, it is often very difficult to work with it as involves a lot of data. It is more beneficial to us to have a global definition that discusses the concepts at the level of categories. For that first notice that an object $c \in \mathcal{C}$ can be regarded as a map $c\colon \Delta^0 \to \mathcal{C}$. So, we can define the category sliced under c as $\mathcal{C}_{c/}$. Now, we have following theorem

Theorem 3.2. *An object $c \in \mathcal{C}$ is initial if $\mathcal{C}_{c/} \to \mathcal{C}$ is a trivial fibration.*

From here on we will employ the second way of thinking about initial objects.

In an ordinary category, initial objects are unique up to unique isomorphism so long as they exist. For quasicategories, the correct analogue is the following.

Fact 3.3. Let $\mathcal{C}^{\text{init}} \subset \mathcal{C}$ be the full subcategory on initial objects of $\mathcal{C}$. Then $\mathcal{C}^{\text{init}}$ is either empty or contractible.

For a proof of this fact see Proposition 1.2.12.9 of [Lu09]. In general you can find a more detailed discussion of initial objects in section 1.2.12.

Colimits

Having gone through all this trouble we are now finally in a position to define colimits in a higher category. Fix a quasicategory $\mathcal{C}$ and a simplicial map $p\colon S \to \mathcal{C}$. Then as promised, we can make the following definition.

Definition 4.1. A *colimit* of p is an initial object of $\mathcal{C}_{p/}$.

In other words, a colimit of p is a vertex $\widetilde{p} \in (\mathcal{C}_{p/})_0$ such that the projection $(\mathcal{C}_{p/})_{\widetilde{p}/} \to (\mathcal{C}_{p/})$ is a trivial fibration. Now, a vertex $\widetilde{p} \in (\mathcal{C}_{p/})_0$ is equivalently a map $\widetilde{p}\colon S^{\triangleright} \to \mathcal{C}$ extending p, and if one has the full definitions, it is not difficult to see we have a diagram

$$\begin{tikzcd} (\mathcal{C}_{p/})_{\widetilde{p}/} \arrow[r] \arrow[d, "\simeq"] & \mathcal{C}_{p/} \arrow[d, equal] \\ \mathcal{C}_{\widetilde{p}/} \arrow[r] & \mathcal{C}_{p/} \end{tikzcd},$$

where the bottom map is restriction along $S \subset S^{\triangleright}$. It is generally easier to deal with the bottom map and forget about taking double slices.

For more details (and some more basic propositions on colimits) see section 1.2.13 of [Lu09].

Our next goal is to discuss some basic features of colimits in higher categories. Then we will use those prove some classic results from classical category theory in this new context.

Restriction along Right Anodyne Maps

An important fact about ordinary colimits is that given a functor $F\colon J \to \mathcal{C}$ and subcategory $I \subset J$, if I is appropriately cofinal in J, then $\operatorname{colim}_J F = \operatorname{colim}_I F$. For example,

$$\operatorname{colim}[X_0 \to X_1 \to X_2 \to \cdots] = \operatorname{colim}[X_{n_1} \to X_{n_2} \to X_{n_3} \to \cdots]$$

for any sequence (n_i) cofinal in $\mathbb{N}$ (i.e. every subsequence of $\mathbb{N}$). This fact has an useful analogue for colimits in quasicategories.

Definition 5.1. Let $\mathrm{RHorn} = \{\Lambda^n_k \subset \Delta^n : 1 \leq n, 1 \leq k \leq n\}$. The saturation of RHorn is the class of *right anodyne* maps.

Example 5.2. $\Lambda^1_1 \subset \Delta^1$ is

$$\begin{tikzcd} 1 & & 1 \\ & \subset & \\ & & 0 \arrow[u] \end{tikzcd}.$$

$\Lambda^2_2 \subset \Delta^2$ is

$$\begin{tikzcd} 1 \arrow[dr] & & 1 \arrow[dr] & \\ & \subset & & \\ 0 \arrow[r] & 2 & 0 \arrow[r] \arrow[uu] & 2 \end{tikzcd}.$$

So, we see that right anodyne maps are an analogue of inclusion of cofinal subdiagrams. Now, the expected fact is the following.

Fact 5.3. If $j\colon S \to T$ is right anodyne, then for any $p\colon T \to \mathcal{C}$, the restriction $\mathcal{C}_{p/} \to \mathcal{C}_{pj/}$ is a trivial fibration.

The proof is not difficult; it is a standard lifting calculus argument. This tells us that certain colimits agree in the following sense. Fix $j\colon S \to T$ right anodyne and $p\colon T \to \mathcal{C}$. Because

$$\begin{tikzcd} T \arrow[r, "j"] \arrow[d] & S \arrow[d] \\ T^{\triangleright} \arrow[r, "\widetilde{j}"] & S^{\triangleright} \end{tikzcd}$$

is a pushout, given $\widetilde{p}\colon S^{\triangleright} \to \mathcal{C}$ extending p,

$$\begin{tikzcd}
\mathcal{C}_{\widetilde{p}/} \arrow[r] \arrow[d] & \mathcal{C}_{\widetilde{p}\widetilde{j}/} \arrow[d] \\
\mathcal{C}_{p/} \arrow[r] & \mathcal{C}_{pj/}
\end{tikzcd}$$

is a pullback, with both top and bottom maps trivial fibrations. As the left and right maps are left fibrations, it follows that one is a trivial fibration if and only if the other is. This tells us that if $\widetilde{p}$ is a colimit if and only if $\widetilde{p}\widetilde{j}$ is.

Example: Pasting Lemma for Pushouts

It is worth giving an example of colimit manipulation. Recall the following lemma, which is a wonderful exercise for those first learning about colimits.

Lemma 6.1. Suppose given a diagram

$$\begin{tikzcd}
1 \arrow[r] \arrow[d] & 2 \arrow[r] \arrow[d] & 3 \arrow[d] \\
4 \arrow[r] & 5 \arrow[r] & 6
\end{tikzcd}$$

in which the left square is a pushout. Then the outer square is a pushout if and only if the right square is. $\square$

We can prove the corresponding statement for pushouts in a quasicategory. First, we must find a way to represent such diagrams in a quasicategory. For this, we can just take the nerve of the pictured category, so such a diagram is just a map $p\colon \Delta^2 \times \Delta^1 \to \mathcal{C}$. This can be pictured as

$$\begin{tikzcd}
 & 2 \arrow[dr] \arrow[dd] \arrow[ddr, shift right] & \\
1 \arrow[ur] \arrow[rr] \arrow[dd] \arrow[ddr, shift left] \arrow[ddrr, shift left] & & 3 \arrow[dd] \\
 & 5 \arrow[dr] & \\
4 \arrow[ur] \arrow[rr] & & 6
\end{tikzcd}.$$

Next, we need to determine what a pushout square is. Recalling that $(\Lambda^2_0)^{\triangleright} = \Delta^1 \times \Delta^1$, we are led to the following definition.

Definition 6.2. Fix a map $\widetilde{q}\colon \Delta^1 \times \Delta^1 \to \mathcal{C}$, and let q denote its restriction to $\Lambda^2_0 \subset \Delta^1 \times \Delta^1$. Then $\widetilde{q}$ is said to be cocartesian if it is a colimit of q, i.e. $\mathcal{C}_{\widetilde{q}/} \to \mathcal{C}_{q/}$ is a trivial fibration.

With this setup, we can prove the following.

Proposition 6.3. Fix $p\colon \Delta^2 \times \Delta^1 \to \mathcal{C}$, pictured as above. For a subet $A \subset \{1,2,3,4,5,6\}$, let $D(A) \subset \Delta^2 \times \Delta^1$ denote the full subcategory on the vertex set A, and let $p(A)$ denote the restriction of p to $D(A)$. Suppose $p(1245)$ is cocartesian. Then $p(1346)$ is cocartesian if and only if $p(2356)$ is.

Proof. We must show that $\mathcal{C}_{p(1346)/} \to \mathcal{C}_{p(134)/}$ is a trivial fibration if and only if $\mathcal{C}_{p(2356)/} \to \mathcal{C}_{p(235)/}$ is. So, consider the diagram

$$\begin{tikzcd}
\mathcal{C}_{p(1346)/} \arrow[d] & \mathcal{C}_{p(123456)/} \arrow[l] \arrow[r] \arrow[d] & \mathcal{C}_{p(2356)/} \arrow[d] \\
\mathcal{C}_{p(134)/} & \mathcal{C}_{p(12345)/} \arrow[l, "\phi"'] \arrow[r, "\psi"] & \mathcal{C}_{p(235)/}
\end{tikzcd},$$

in which all vertical maps are fibrations. Notice that both squares are pullbacks, as both squares in

$$\begin{tikzcd}
D(134) \arrow[r] \arrow[d] & D(12345) \arrow[d] & D(235) \arrow[l] \arrow[d] \\
D(1346) \arrow[r] & D(123456) & D(2356) \arrow[l]
\end{tikzcd}$$

are pushouts. We will show that ϕ and ψ are trivial fibrations. It follows that the left map is a trivial fibration if and only if the right map is, which is what we wish to show. First, $D(235) \subset D(12345)$, pictured as

$$\begin{tikzcd}
2 \arrow[r] \arrow[d] & 3 \\
5 & \\
& \subset & \\
1 \arrow[r] \arrow[d] \arrow[dr] \arrow[rr, bend left] & 2 \arrow[r] \arrow[d] & 3 \\
4 \arrow[r] & 5 &
\end{tikzcd},$$

is easily seen to be a right anodyne map, so that ψ is a trivial fibration. Next, notice that ϕ factors as $\mathcal{C}_{p(12345)/} \to \mathcal{C}_{p(1234)/} \to \mathcal{C}_{p(134)/}$. As $D(134) \subset D(1234)$, pictured as

$$\begin{tikzcd}
1 \arrow[d] \arrow[rr, bend left] & & 3 \\
4 & & \\
& \subset & \\
1 \arrow[r] \arrow[d] \arrow[rr, bend left] & 2 \arrow[r] & 3 \\
4 & &
\end{tikzcd},$$

is right anodyne, the second map is a trivial fibration. All that remains is the map $\mathcal{C}_{p(12345)/} \to \mathcal{C}_{p(1234)/}$. But

$$\begin{tikzcd} D(124) \arrow[r] \arrow[d] & D(1234) \arrow[d] \\ D(1245) \arrow[r] & D(12345) \end{tikzcd}$$

is a pushout, so

$$\begin{tikzcd} \mathcal{C}_{p(12345)/} \arrow[r] \arrow[d] & \mathcal{C}_{p(1245)/} \arrow[d] \\ \mathcal{C}_{p(1234)/} \arrow[r] & \mathcal{C}_{p(124)/} \end{tikzcd}$$

is a pullback. Because $p(1245)$ is cocartesian, the right map is a trivial fibration, whence so too is the left map. This completes the proof. $\square$

Exercise: Mapping Space and Colimits

If $F\colon I \to \mathcal{C}$ is a functor of ordinary categories, we have $\mathcal{C}(\operatorname{colim}_i F(i), c) \cong \lim_i \mathcal{C}(F(i), c)$. An analogue is true for colimits in a quasicategory.

Fact 7.1. For any simplicial set S, the inclusion $\{v\} \to S^{\triangleright}$ of the cone point is right anodyne.

Definition 7.2. For $x, y \in \mathcal{C}$, define the *right mapping space* $\operatorname{map}^R_{\mathcal{C}}(x, y)$ by the pullback

$$\begin{tikzcd} \operatorname{map}^R_{\mathcal{C}}(x,y) \arrow[r] \arrow[d] & \mathcal{C}_{x/} \arrow[d] \\ \Delta^0 \arrow[r, "y"] & \mathcal{C} \end{tikzcd} .$$

It turns out that the right mapping space is a model for any correctly defined mapping space. Using these, the following is left as an exercise.

Exercise 7.3. Suppose $p\colon S \to \mathcal{C}$ has colimit $\widetilde{p} = p * v\colon S * \Delta^0 \to \mathcal{C}$. Define P_y by the pullback

$$\begin{tikzcd} P \arrow[r] \arrow[d] & \mathcal{C}_{p/} \arrow[d] \\ \Delta^0 \arrow[r, "y"] & \mathcal{C} \end{tikzcd} .$$

Then $P_y \cong \operatorname{map}^R_{\mathcal{C}}(v, y)$.

Functoriality of Colimits

If $\mathcal{C}$ is an ordinary category that admits all small colimits and J is a small category, then we can assemble these colimits into a functor $\operatorname{colim}\colon \operatorname{Fun}(J, \mathcal{C}) \to \mathcal{C}$. This is, for instance, what tells us that a diagram like

$$\begin{tikzcd} X(1) \arrow[d] & X(0) \arrow[l] \arrow[r] \arrow[d] & X(2) \arrow[d] \\ Y(1) & Y(0) \arrow[l] \arrow[r] & Y(2) \end{tikzcd}$$

induces a map

$$X(1) \cup_{X(0)} X(2) \to Y(1) \cup_{Y(0)} Y(2).$$

However, the colimit of a functor is not uniquely defined, but rather defined up to unique isomorphism. In other words, there is a contractible groupoid worth of choices. Of course, this discussion still holds with easy modifications if $\mathcal{C}$ only admits certain colimits.

All of this upgrades to colimits in quasicategories. The expected upgrade should look something like: for a quasicategory $\mathcal{C}$ and simplicial set S, there is a functor $\operatorname{colim} \operatorname{Fun}^{\exists\mathrm{colim}}(S, \mathcal{C}) \to \mathcal{C}$, and this functor is defined up to contractible choice. Formally, what's true is the following.

Fact 8.1. In the diagram

$$\begin{tikzcd} \operatorname{Fun}^{\mathrm{colim\ cone}}(S^{\triangleright}, \mathcal{C}) \arrow[r] \arrow[d, "q"] & \operatorname{Fun}(S^{\triangleright}, \mathcal{C}) \arrow[d] \\ \operatorname{Fun}^{\exists\mathrm{colim}}(S, \mathcal{C}) \arrow[r] & \operatorname{Fun}(S, \mathcal{C}) \end{tikzcd},$$

the left map q is a trivial fibration.

It follows that there is a contractible space of sections of q, and composing with the evaluation map gives the desired colimit functor.

For a proof of that see 50.5 in [Re16].

References

[Lu09] J. Lurie, *Higher Topos Theory*, Annals of Mathematics Studies 170, Princeton University Press, Princeton, NJ, 2009, xviii+925 pp. A

[Re16] C. Rezk, *Stuff About Quasicategories* Unpublished notes, http://www.math.uiuc.edu/ rezk/595-fal16/quasicats.pdf